

Multi-Chamber Camera Bioreactor (MCCB)

MAE 156B - Final Sponsor Presentation

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Problem Statement Overview

Our Task:

To develop a bioreactor that can apply following external stimuli to stem cells:

- Electric Field
- Magnetic Field
- Fluid Input/Output

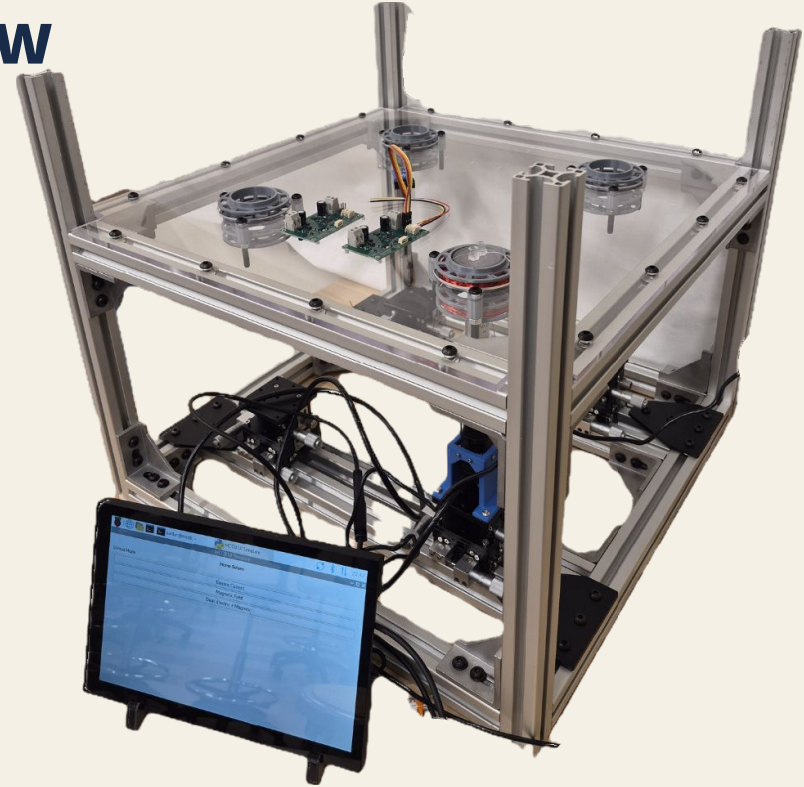
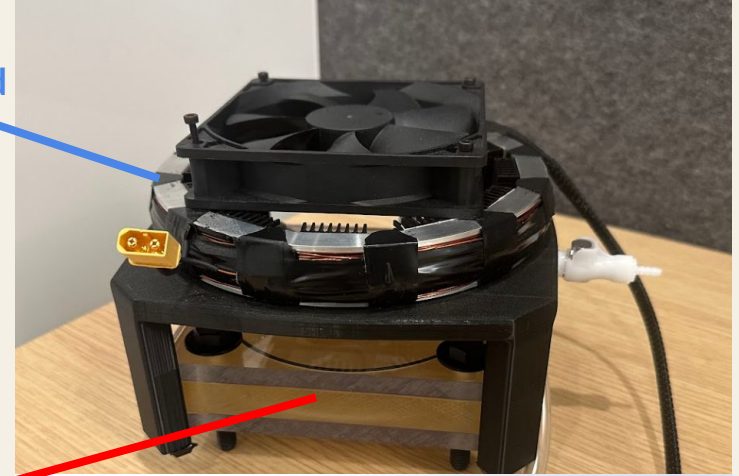


Figure 2 - Photo of the MCCB

Previous Design - 2019

- **Successes:**
 - Successfully created magnetic field generator
 - Leak free chamber allowing for fluid flow
 - Platinum wire electrodes inside chamber to apply electric field
- **Requested Design Improvements**
 - Multiple chambers
 - Individualized Stimuli control per chamber
 - Incorporated Microscope Imaging

Magnetic Field Generator



Single Chamber

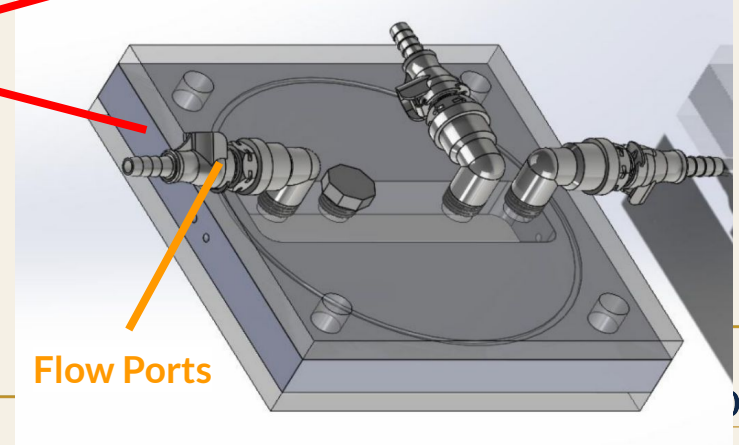
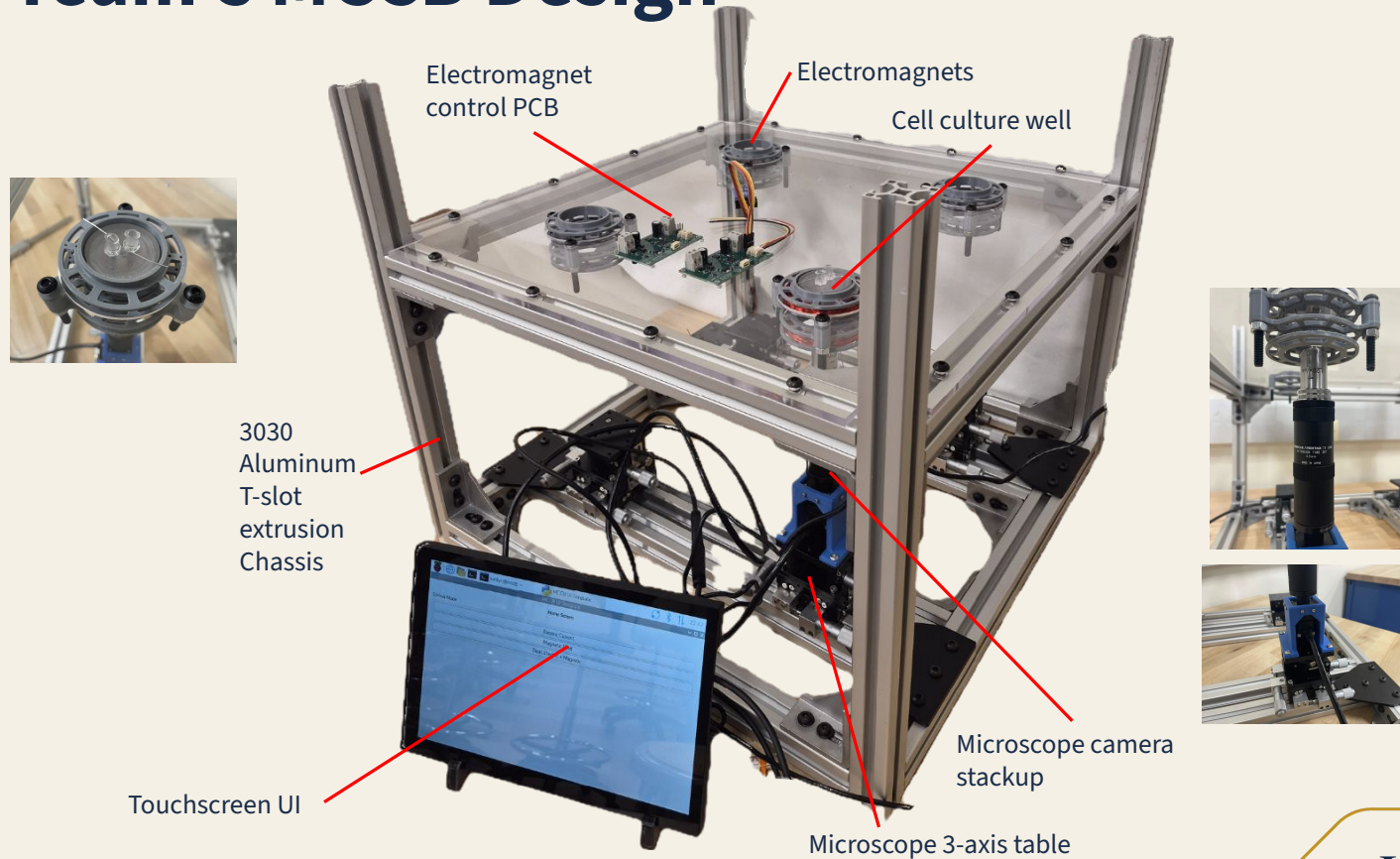


Figure 3 - 2019
MAE 156 Stimuli
Chamber

Flow Ports

Our Team's MCCB Design



Integrated MCCB System Block Diagram

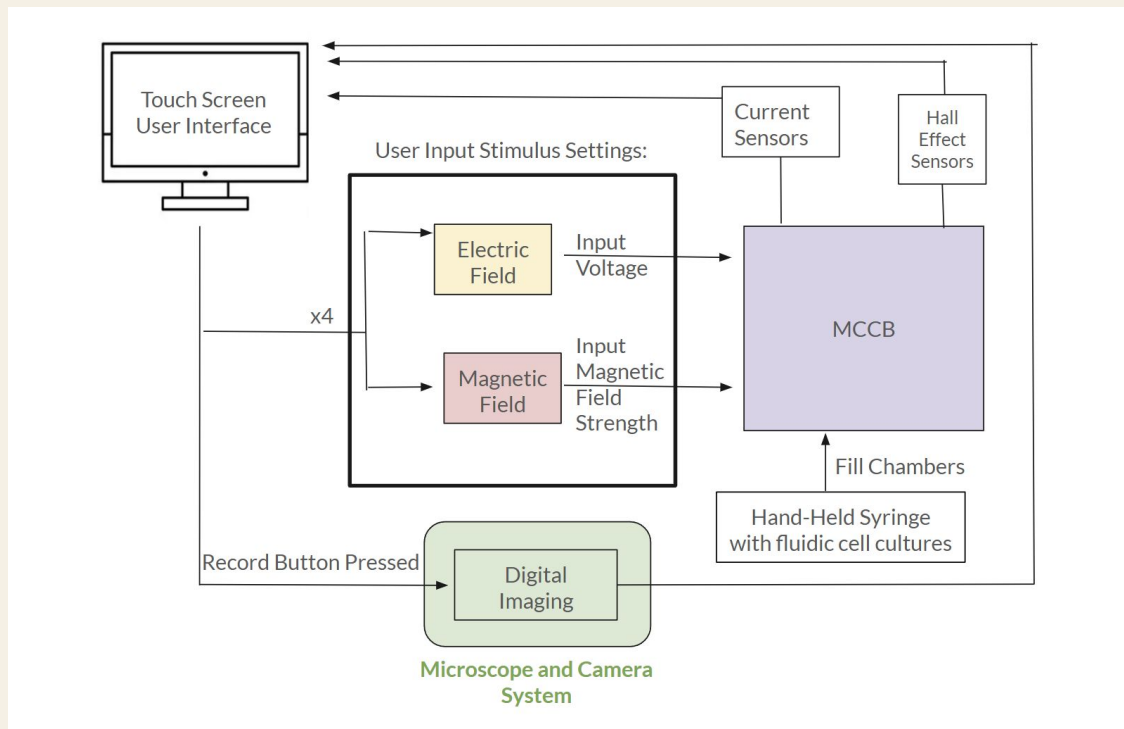
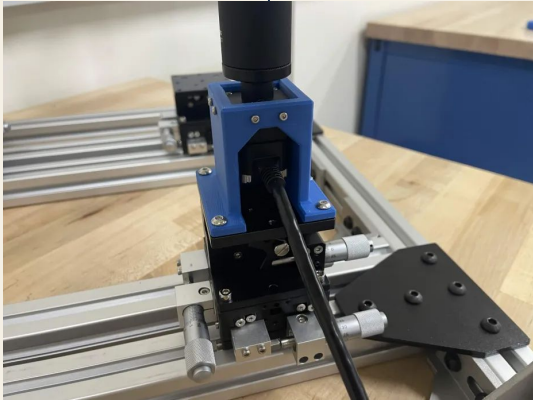


Figure 5 - Block Diagram of the Full Integrated System

Single Well System Breakdown

a)



b)



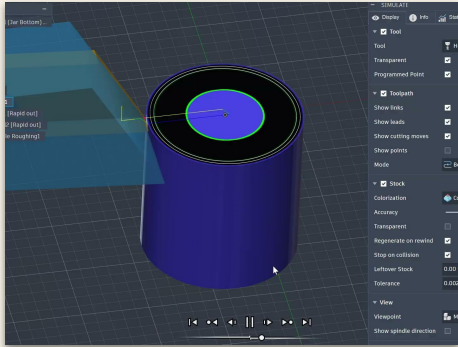
c)



Figure 6 - (a) photo of the microscope mounting (b) figure of the microscope underneath the well (c) top view of the well system

Well Design Iterations

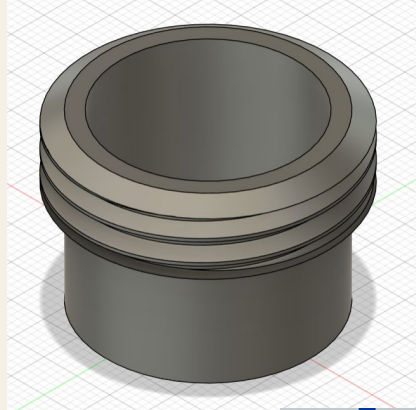
Off-the Market Jars



- Fastest
production



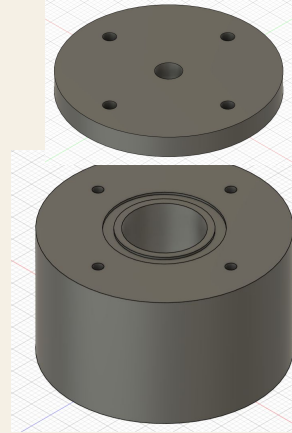
Machined Delrin (screw top)



- most accessible
fabrication



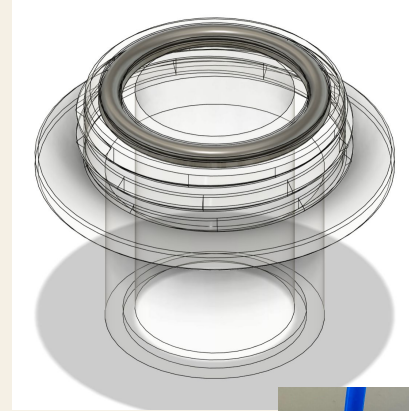
Bolted acrylic lid, machined delrin body



- Best
Clamping



VeroClear Resin 3D Print



-Most
Customizable



Final Well Design

Assembly Breakdown

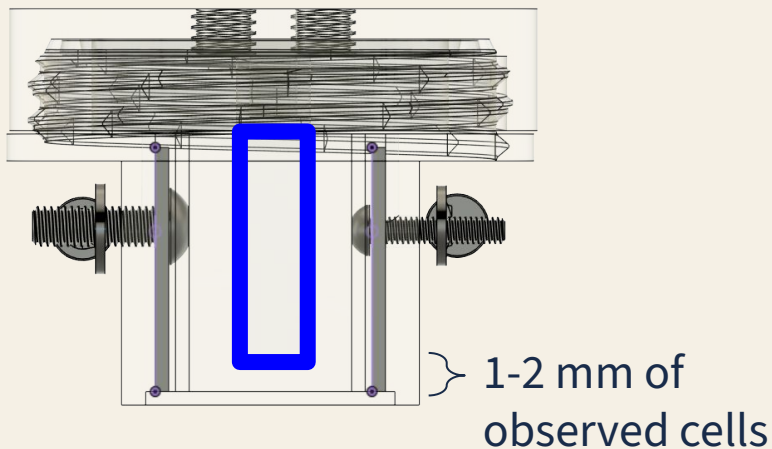


Figure 8.a - Side View CAD of the Well highlighting the plunger



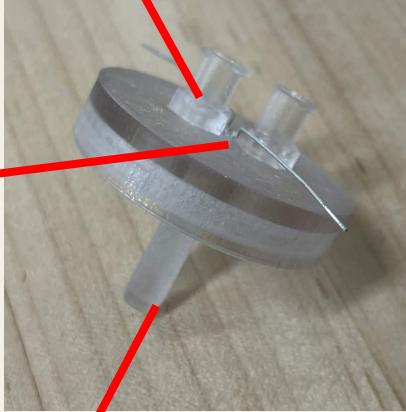
Figure 8.b - Semi-exploded Well CAD

Component Breakdown

Luer Locks

LED

Diffusive
Backing



Ring terminal and bolt

Titanium Electrode

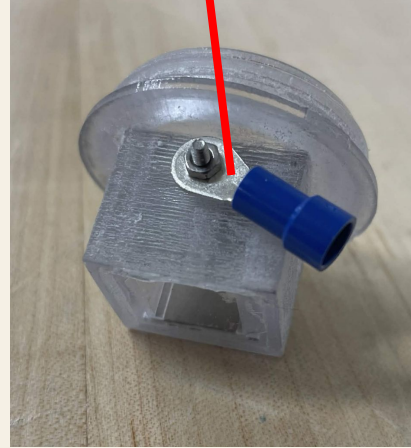


Figure 9 - Labeled Photos of the Well

Electrodes

Significant Design Decisions

Configuration	Pros	Cons
Square Plates	Most uniform field	More Material Needed
Straight Wires to Bottom	Easy to Secure to Lid Easiest Fabrication	Field Concentrated around Wires

Electrode
Configuration

Material
Selection

Fine-tuning
Incorporation into
the System

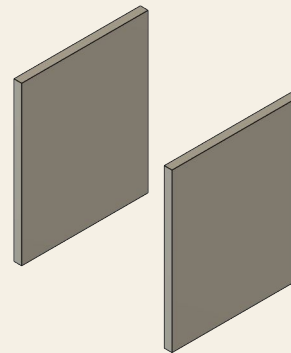


Figure 10.a Parallel Plate Electrodes

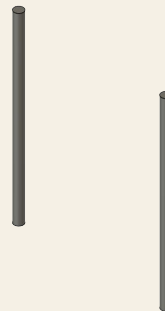


Figure 10.b Parallel Wire Electrodes

Electrodes

Electrode Material and Configuration Selection

(Point Values Assigned out of 5)

Material	Biocompatibility	Durability	Corrosion Resistance	Approximate Cost (per gram)
Titanium	5	5	4	\$0.03
Gold	4	3	5	\$145
Platinum	5	4	4	\$70

Electrodes

Fastener Details

Design Option	Pros	Cons
Leaded Solder	Readily available, easy assembly	Does not adhere to TiO, difficult assembly
Conductive Epoxy	Does not toxify saline	Does not adhere to TiO
Wrapped wire	Simple design, no extra parts	Poor/inconsistent contact to plate
Bolted ring terminal	Industry standard crimping electrical contact	High part count

Electrode Control Validation Tests

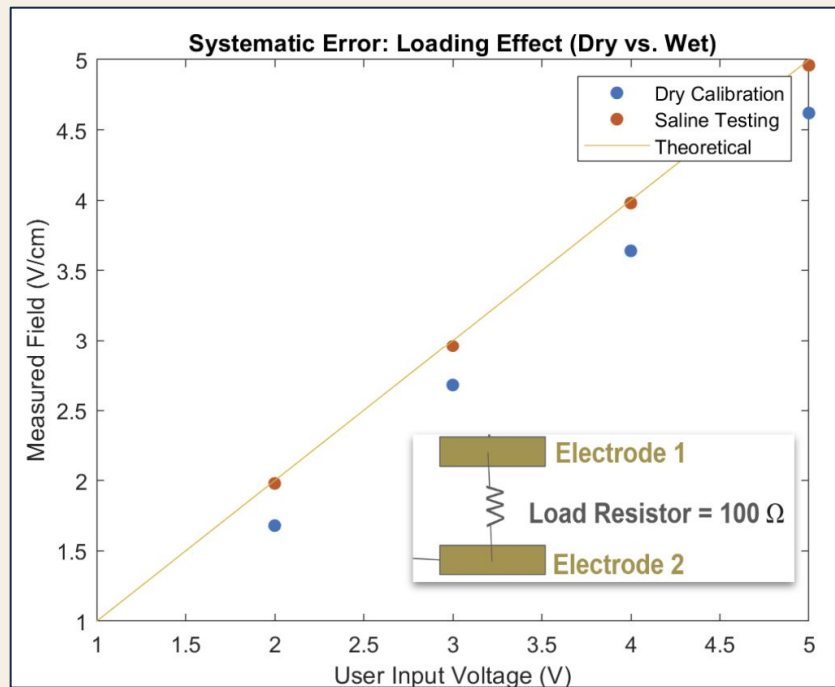


Figure 13.a - Measured drive voltage at the electrodes against input voltage. Dry environment tests with a load resistor to approximate the resistance of saline was used for control and sensor calibration. Submerged saline solution tests exhibited no significant loss through the contacts

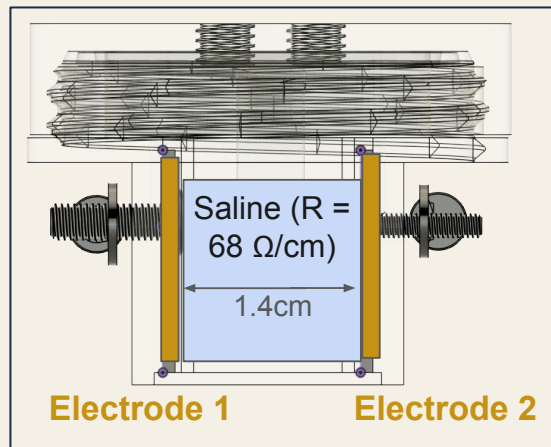


Figure 13.b Electrode interaction with the saline solution that has an equivalent resistance of 95.2 ohms across a 1.4 cm gap

$$R_{\text{saline gap}} = 68 \frac{\Omega}{\text{cm}} \cdot 1.4 \text{ cm}$$

$$= 95.2 \Omega \approx 100 \Omega$$

Load resistor selected for dry environment calibration test

→ Compare to tests in saline

- PWM control for adjustable voltage load
- INA219 current sensor and multimeter used to verify expected current draw and voltage at the electrodes ($V=IR$)

Electrode Control - Function Generation

- AC Control requested by sponsor (1-100 Hz)
- ESP32 PWM control to H-bridge (Pololu DRV8876)
- RC filter required for Sine wave to appear on oscilloscope

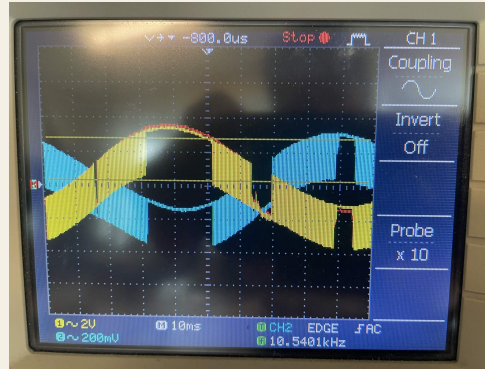


Figure 14 a - Sine wave generated via PWM WITH RC filter

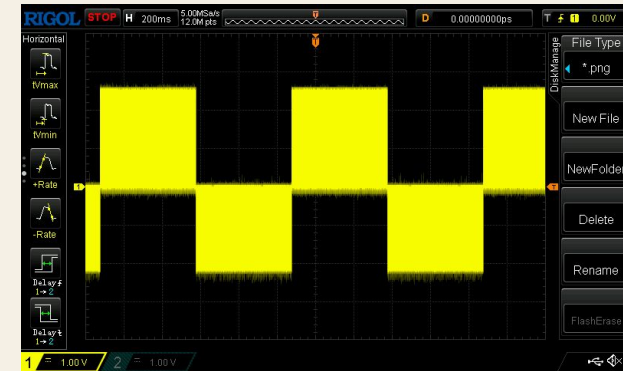
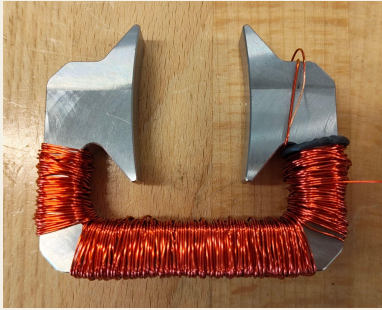
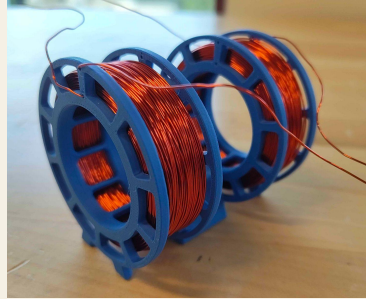


Figure 14.b - Generated Square wave

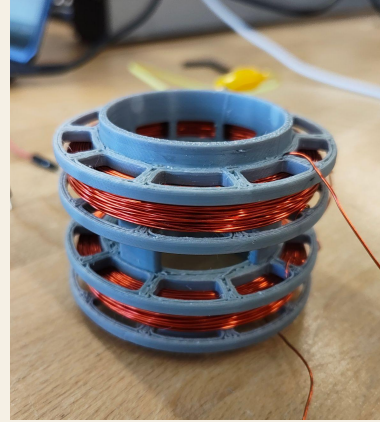
Electromagnet Design Iterations



Horseshoe Magnet



Horizontal Mount
Helmholtz Coil



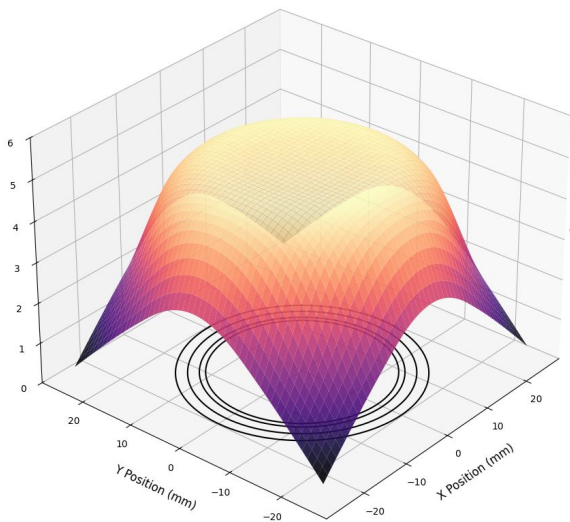
Vertical Mount
Connected
Helmholtz Coil



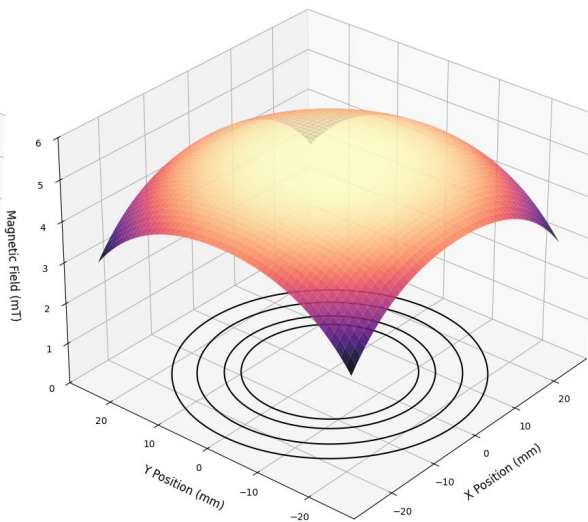
Final Version

Magnetic Field: Uniformity Validation

Measured 3D Surface



Theoretical 3D Surface



Methods:

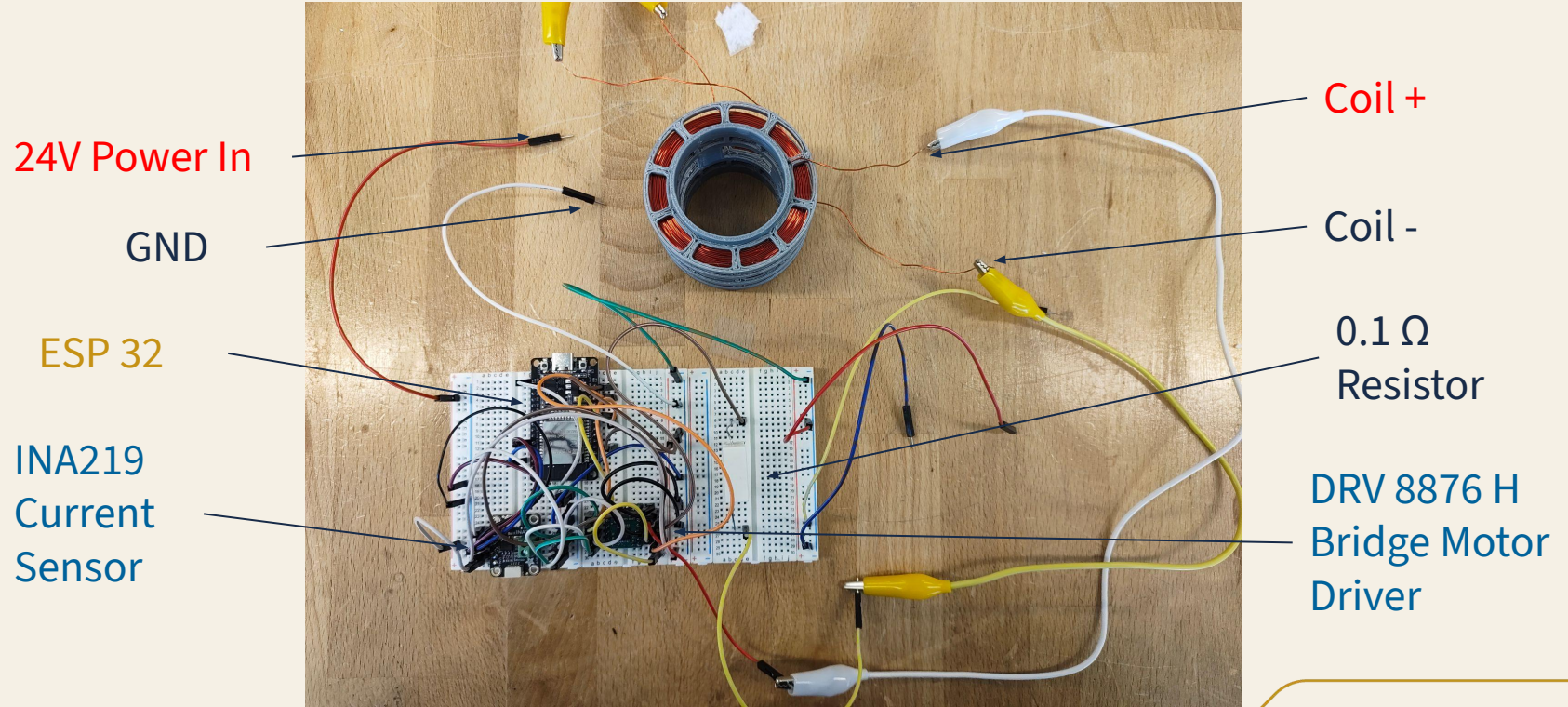
- Magnetometer sweep across XY and Z planes
- 0.2 mT offset applied to remove Earth's background field
- Values radially interpolated (full data in Appendix)

Results:

- Measured surface closely tracks theoretical model
- Field drop-off at edges slightly steeper than theory
- Entire well sits within the 99% uniformity contour (<1% deviation from center)
- Z-axis variation stays minimal within ± 10 mm of coil face-to-face

Figure 16.a - Graphs of measured field vs theoretical with contour lines for 99%, 98%, 95%, and 90% regions of uniformity.

Electromagnet Driving Circuit (Breadboard Prototype)



Custom Helmholtz Coil PCBs

Hall Effect Sensor

- Used in the final design
- Allows for closed loop control

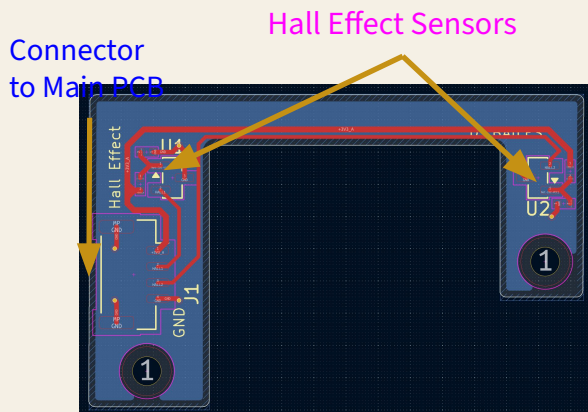


Figure 18.a: Hall Effect Satellite Board

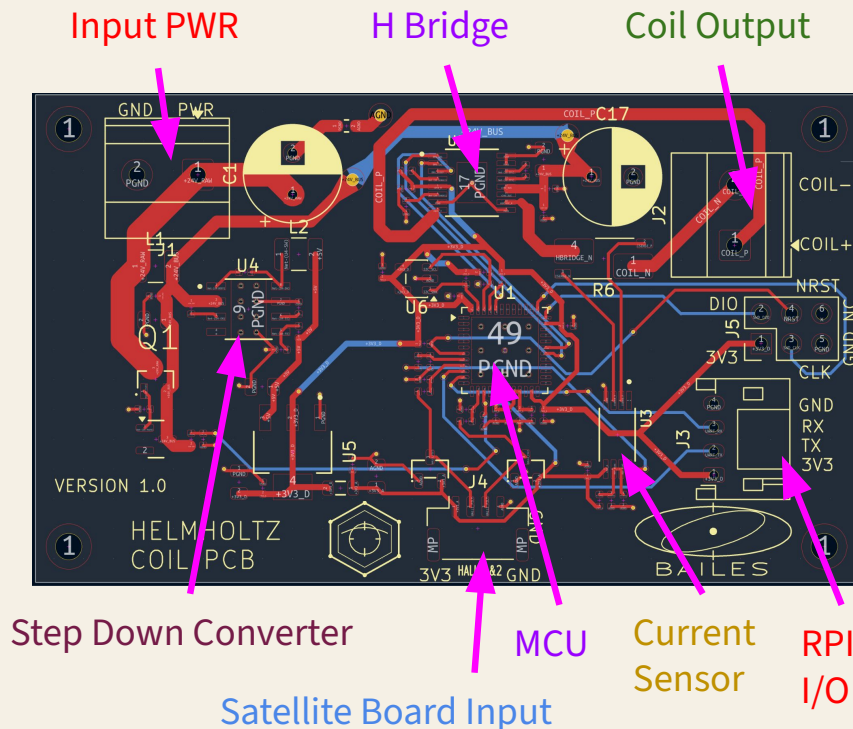
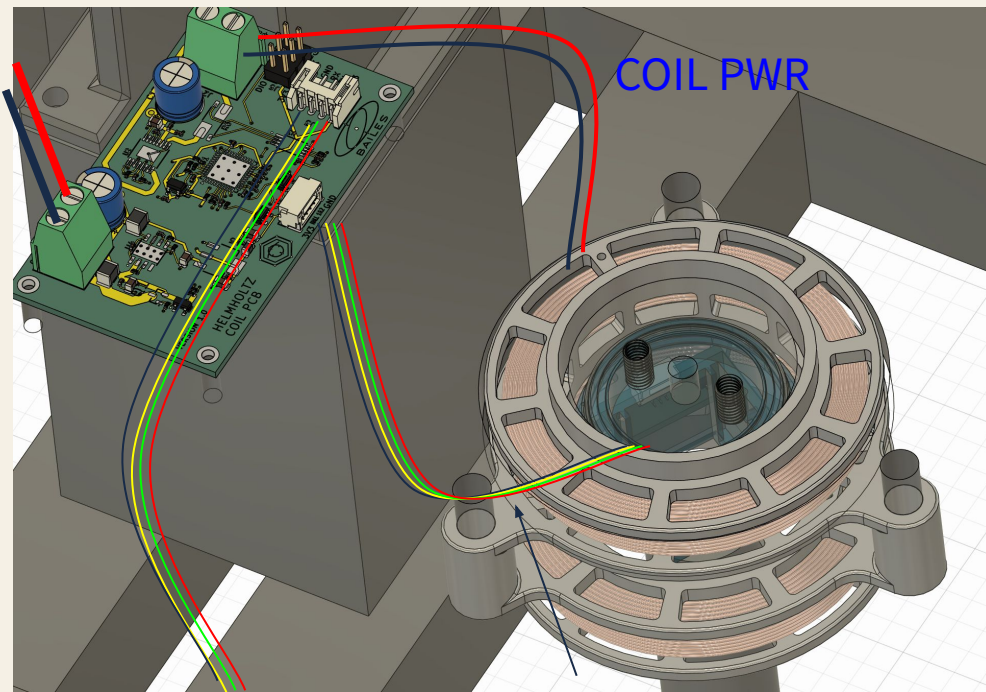


Figure 18.b - Helmholtz Coil PCB
Design V1 - scrapped

Electromagnet Subsystem: Ideal Integration Summary

PWR INPUT

- **Mechanical:** ASA housing, well mount, PCB mount, ring terminal integration
- **Electrical:** STM32 + H-Bridge, 24V/3A, 0–250 Hz AC/DC, PID closed loop
- **Validation:** <1% field deviation across full well area, Z-variance minimal ± 10 mm, sensor reads at 1000 Hz



COIL PWR

COMMS to RPI

SATELLITE
CABLE

Figure 19a: CAD of PCB and well housing Helmholtz Coil integration

Helmholtz Coil Output Data

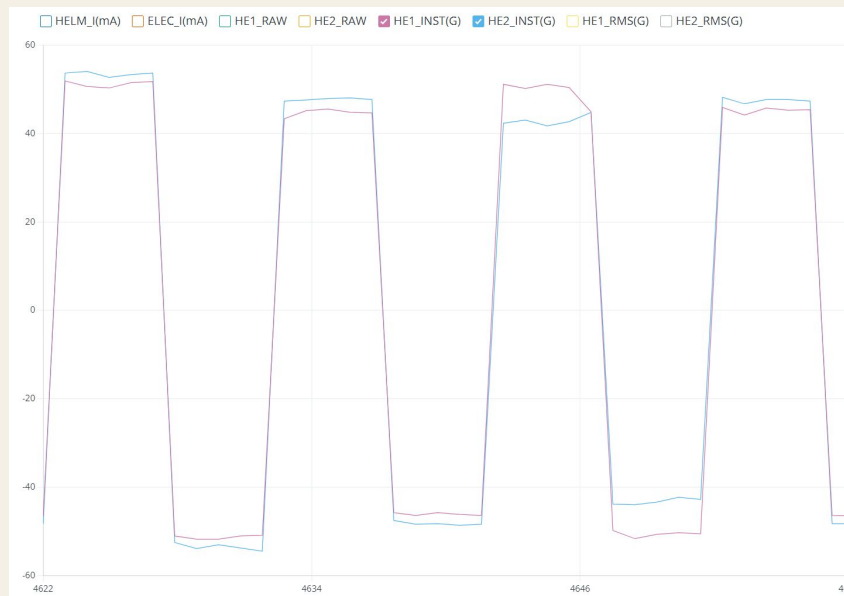


Figure 20 - square wave of the electromagnet pwm (1 Hz)

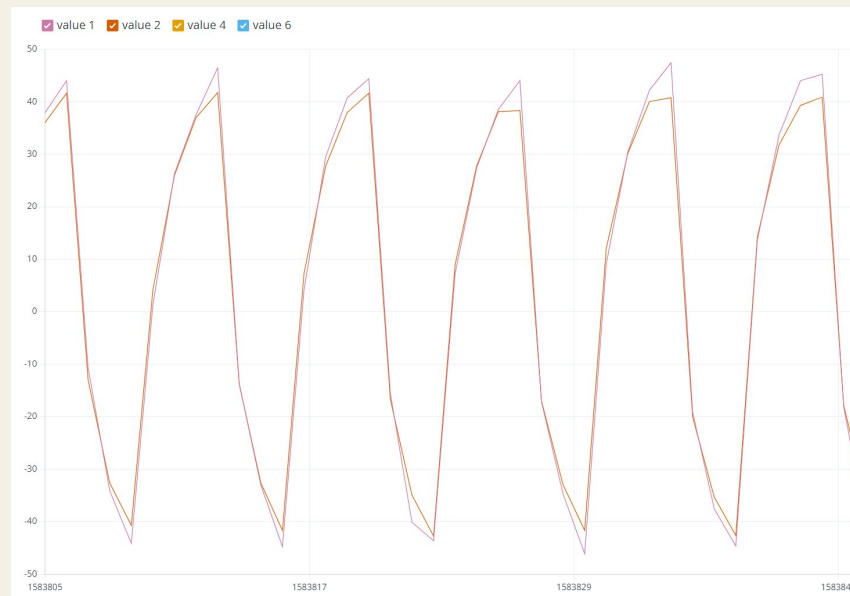


Figure 20 - square wave of the electromagnet pwm (250 Hz)

Optical Train Components

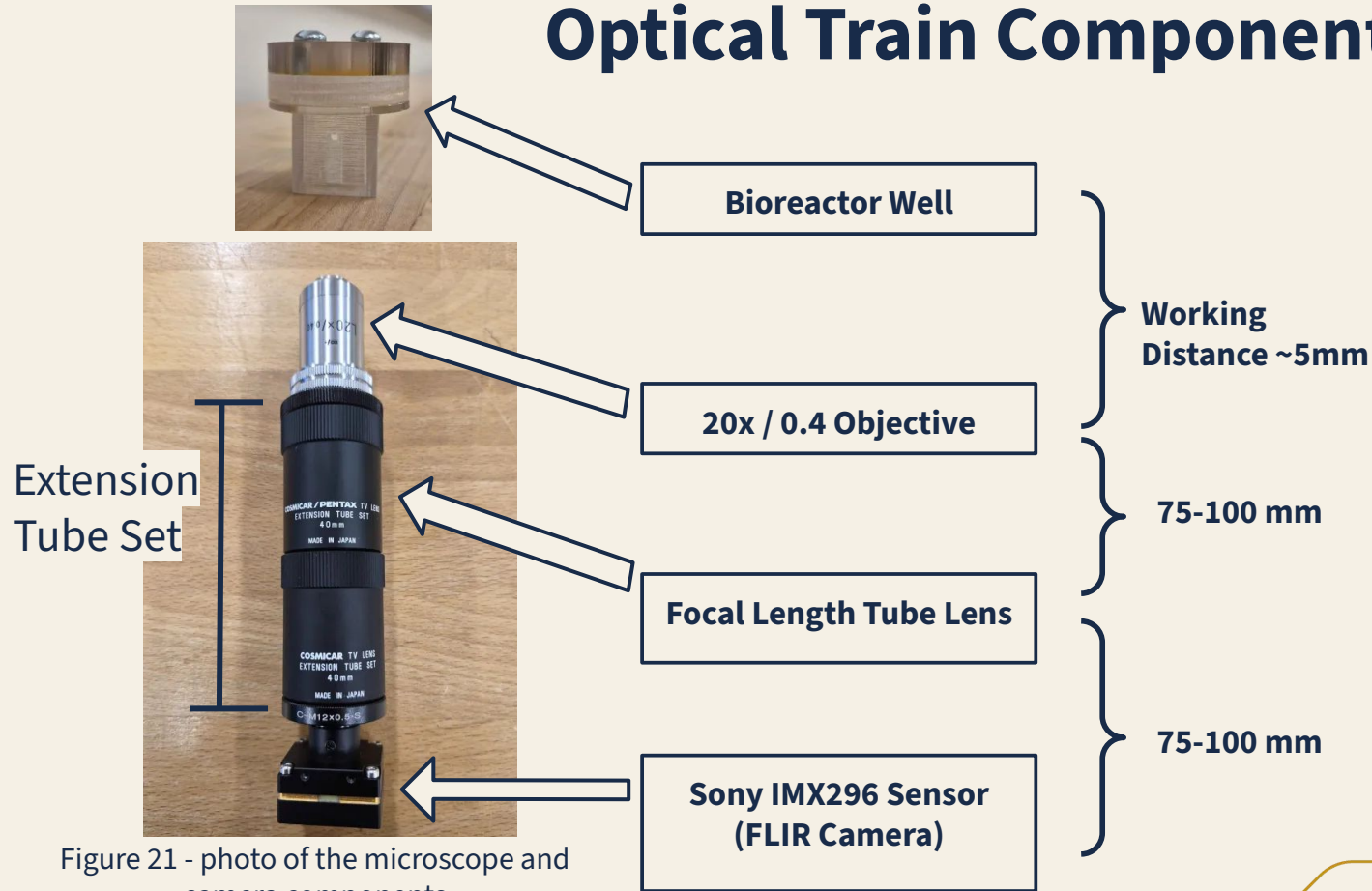


Figure 21 - photo of the microscope and camera components

Optics - Reticle Lens Resolution Test

Goal: Image zebrafish embryos (~25 microns in diameter)

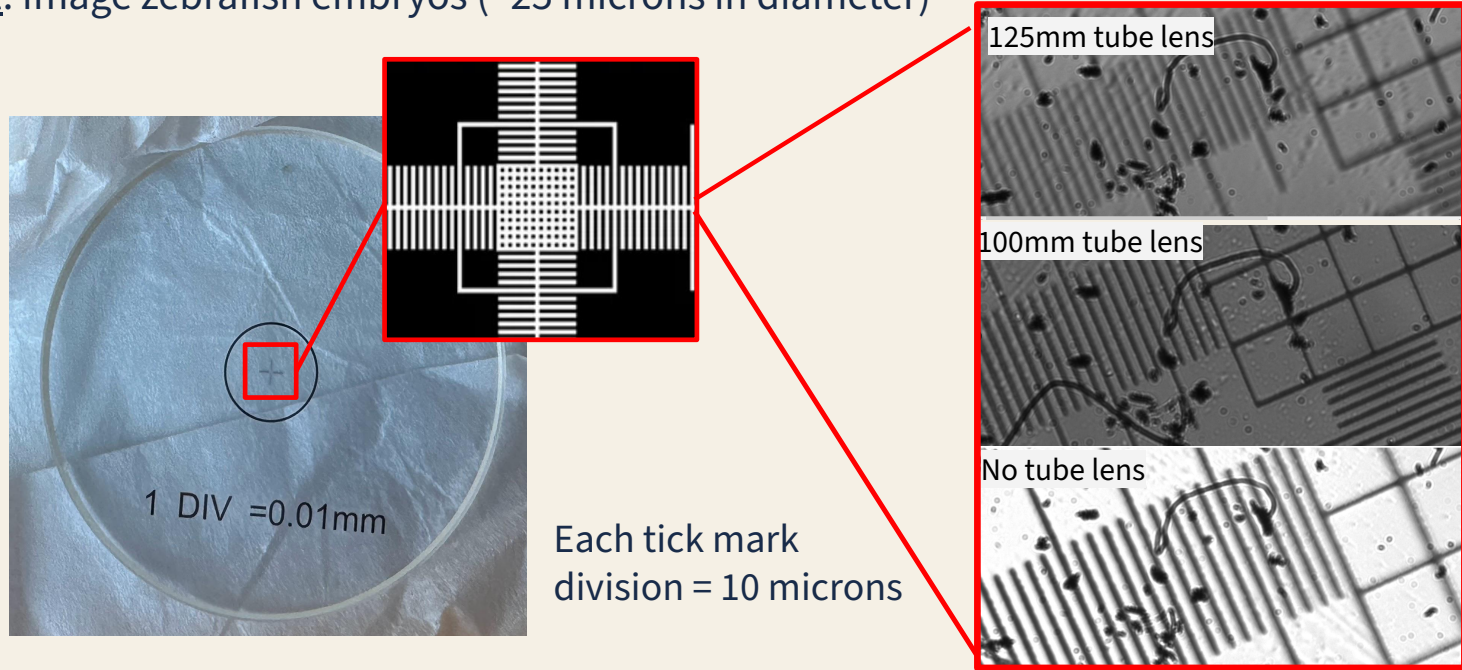


Figure 22 - Reticle Lens Tick Mark Image Tests With Varying Tube Lens Focal Lengths

Fixed Cell Sample Imaging Tests



Fig 23a. Scanning Environment With Microscope



Fig 23b. Mysterious Particle Detected

Central Electronics

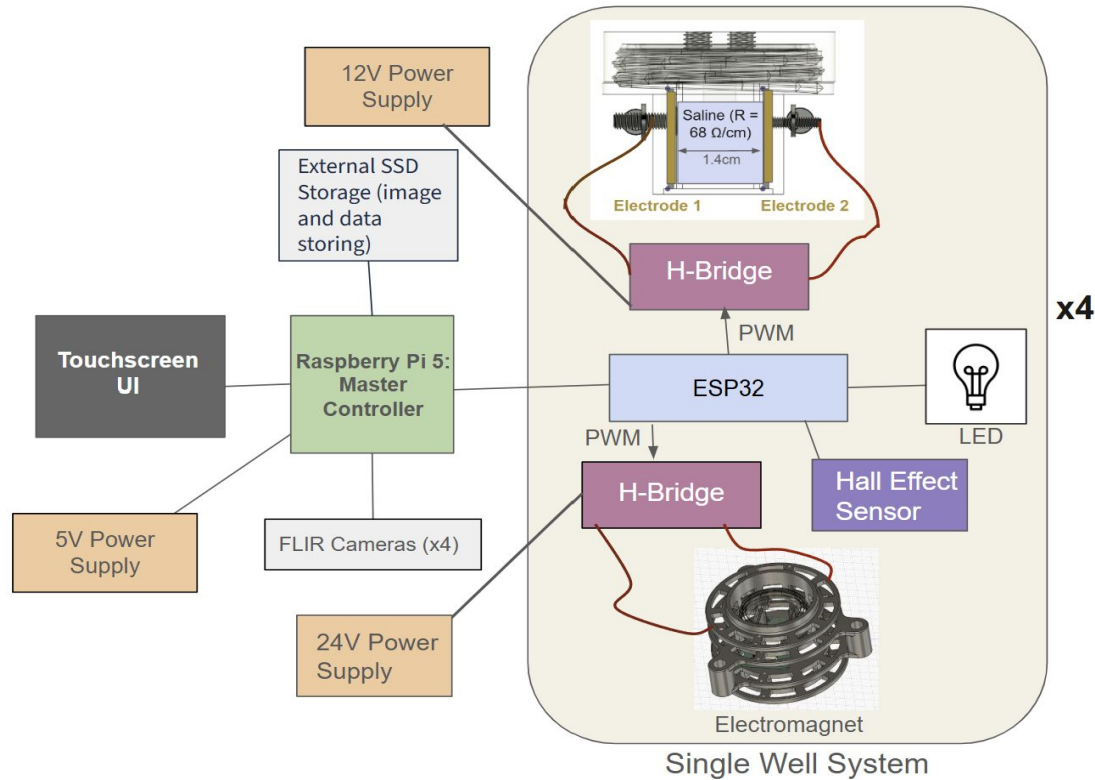
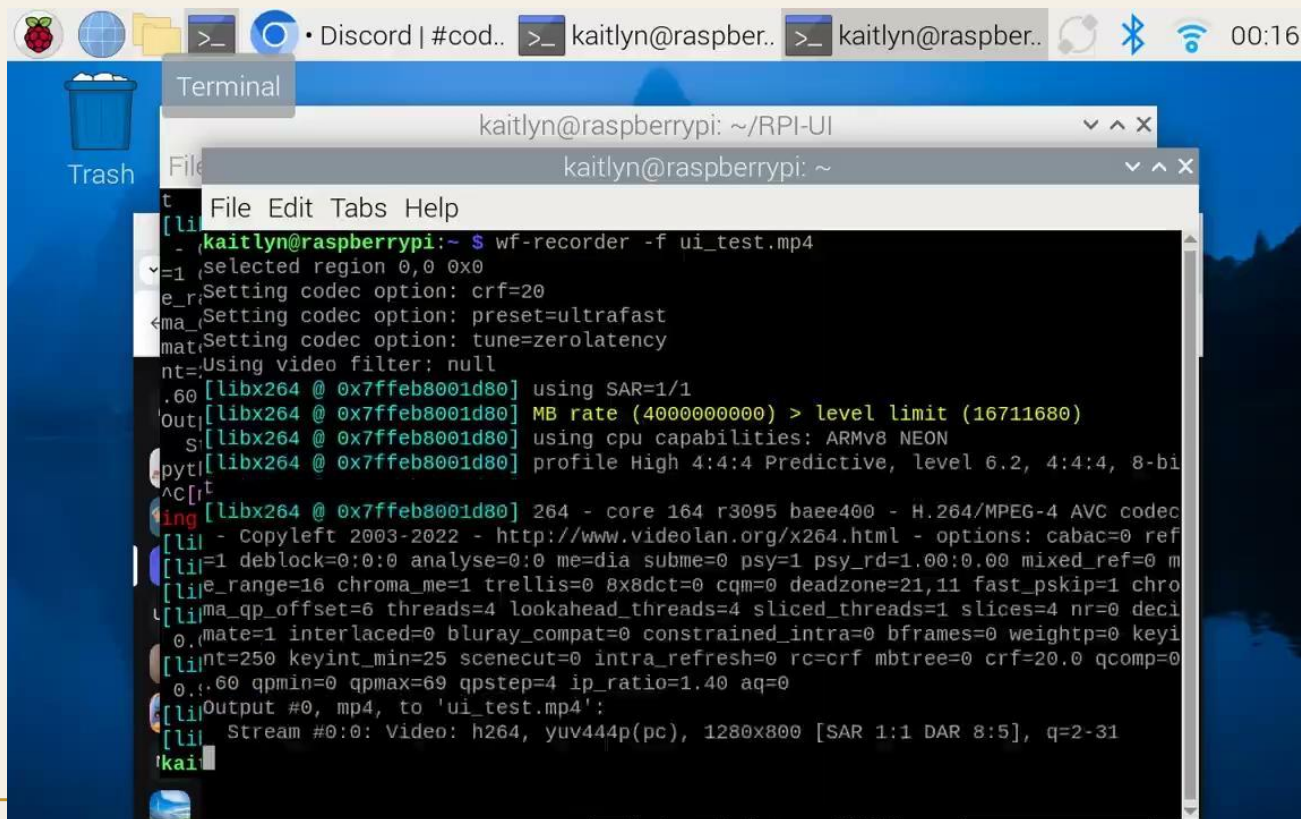


Fig. 24 - High Level Control System Layout

User Interface - Video Demonstration



User Interface - USB Assignment

01. CONNECTION

ASSIGN SERIAL DEVICES TO WELLS

WELL 1:

(NONE)

/dev/ttyAMA10

WELL 2:

(NONE)

WELL 3:

(NONE)

WELL 4:

(NONE)

Each well should have its own ESP32 device.
ESP32s typically appear as CP210x or CH340.

↻ REFRESH PORTS

EXIT APPLICATION

CONNECT CONTINUE

UC San Diego

User Interface - Home Screen

MCCB CONTROLLER					RECONFIGURE PORTS	EXIT APPLICATION
CONTROL	WELL 1	WELL 2	WELL 3	WELL 4	IMAGING	

02. STIMULATION

MCCB STIMULATION CONTROL

SELECT MODE

ELECTRIC FIELD

MAGNETIC FIELD

DUAL: ELECTRIC + MAGNETIC

User Interface - Stimuli Control

03. PARAMETERS

DUAL: ELECTRIC + MAGNETIC

WELL 1

ELECTRIC (V/CM): 0 - 1.5 V/CM

MAGNETIC (GAUSS): 0 - 15.0 G...

WELL 2

ELECTRIC (V/CM): 0 - 1.5 V/CM

MAGNETIC (GAUSS): 0 - 15.0 G...

WELL 3

ELECTRIC (V/CM): 0 - 1.5 V/CM

MAGNETIC (GAUSS): 0 - 15.0 G...

WELL 4

ELECTRIC (V/CM): 0 - 1.5 V/CM

MAGNETIC (GAUSS): 0 - 15.0 G...

TAP A FIELD TO EDIT

7	8	9
4	5	6
1	2	3
.	0	⌫

CLEAR CONFIRM

BACK APPLY PARAMETERS

User Interface

MCCB CONTROLLER					RECONFIGURE PORTS		EXIT APPLICATION	
CONTROL	WELL 1	WELL 2	WELL 3	WELL 4	IMAGING			
WELL 1		WELL 1 // ALL DATA					PAUSE	CLEAR
ALL DATA								
ELECTRIC								
MAGNETIC								

User Interface

MCCB CONTROLLER					RECONFIGURE PORTS	EXIT APPLICATION
CONTROL	WELL 1	WELL 2	WELL 3	WELL 4	IMAGING	
WELL 01 STOPPED		WELL 02 STOPPED		CAMERA SETTINGS		
				EXPOSURE		
				5 ms		
				GAIN		
				0 dB		
				FRAME RATE		
				10 FPS		
WELL 03 STOPPED		WELL 04 STOPPED		APPLY SETTINGS		
				APPLY TO ALL		
				SNAPSHOTS		

Conclusions

- **Integrated Research Platform:** Engineered a fully integrated 4-well bioreactor enabling simultaneous electromagnetic stimulation and real-time microscopic observation.
- **Precise Stimuli Delivery:** Applies highly uniform electric (0–1.5 V/cm via titanium electrodes) and magnetic (0–50 Gauss via Helmholtz coils) fields to cell cultures.
- **Sterile & Real-Time Monitoring:** Features a Raspberry Pi touchscreen UI and integrated imaging software to monitor cells and validate fields (via current and Hall effect sensors) without breaking the sterile seal.
- **Impact on Regenerative Medicine:** Provides our sponsor, Dr. Peter Chen, with a specialized tool to conduct highly reproducible stem cell differentiation trials.

Future Design Recommendations

Biocompatible Resin

- Removes the need for sterilization with sodium hydroxide
- Will eliminate the risk of resin leaching and toxifying the sample

Improve Optics

- Try different eyepiece magnifications, objectives, etc.
- Coat lid of well so resin does not interfere with image view

Platinum Electrodes

- Bioengineering standard to prevent corrosion
- Prevents buildup of resistive titanium oxide layer

Printed Circuit Board Incorporation

- More organized electronics setup

Downsize System Footprint

- Decrease overall size

Incorporate Hydrostatic Pressure Stimuli

- Risk reduction for well design showed promising results for holding pressure (See appendix for more information)

Thank you

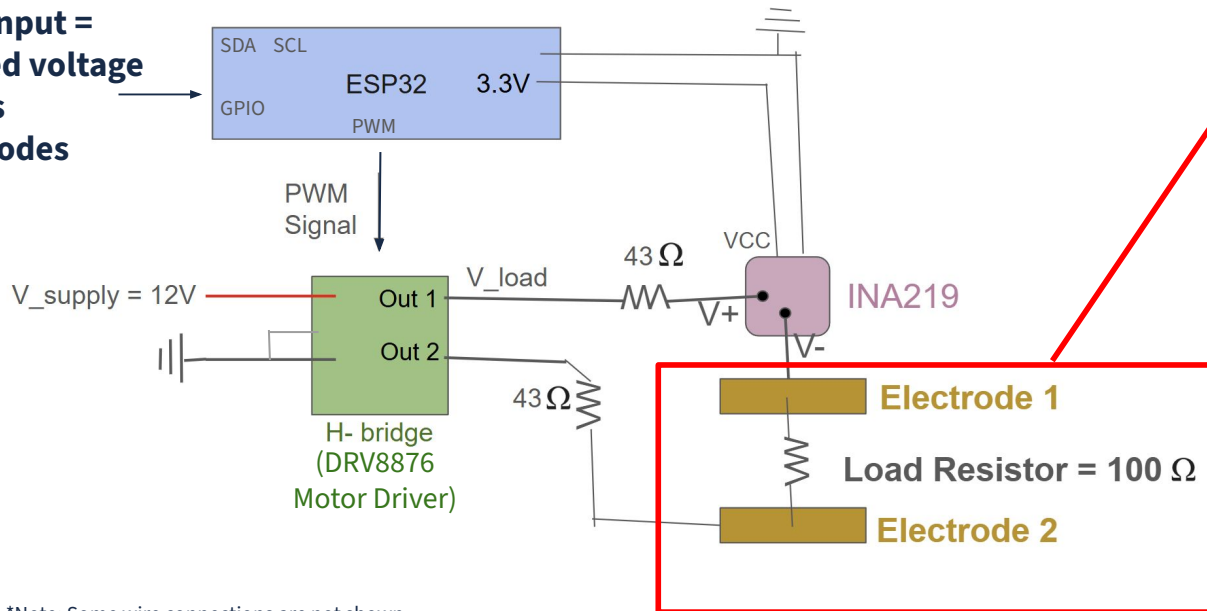
Appendix Slides

Electrode Testing and Validation - Dry Electrode Test

Control Calibration

- Electrode Control FR: Establish a controllable AC Electric Field, adjustable from 2-5 V/cm

User input =
desired voltage
across
electrodes



*Note: Some wire connections are not shown

Load resistor approximates resistance of saline solution across a 1.4 cm gap between electrodes

$$R_{\text{saline}} = 68 \frac{\Omega}{\text{cm}}$$

$$R_{\text{saline gap}} = 68 \frac{\Omega}{\text{cm}} \cdot 1.4 \text{ cm}$$
$$= 95.2 \Omega \approx 100 \Omega$$

Fig. 10 - Single Well Electrode Circuit Schematic

Control Calibration



Compare with Multimeter → Calibrate PWM duty cycle and INA219 sensor

Eq 11a. expected current/draw: $I_{expected} = (\Delta V_{target}/R_{saline})$

Power: 122.00 mW

INA219 Readings

expected voltage of electrodes: 2.25V

Fig 11b. Serial Monitor Output

$$V_{electrode} = V_{bus} - (\frac{V_{supply}}{R1+R+R_{Load}} \cdot R2)$$

Eq 11b. expected electrode voltage

Electrode Testing and Validation - Saline Solution Tests

System Integration Test

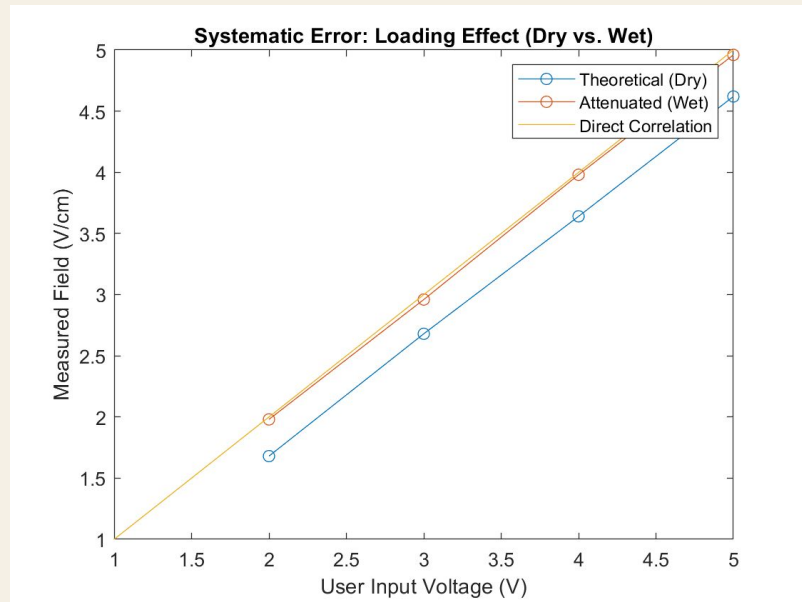


Figure 12.a - Measured Voltage vs User Input Voltage

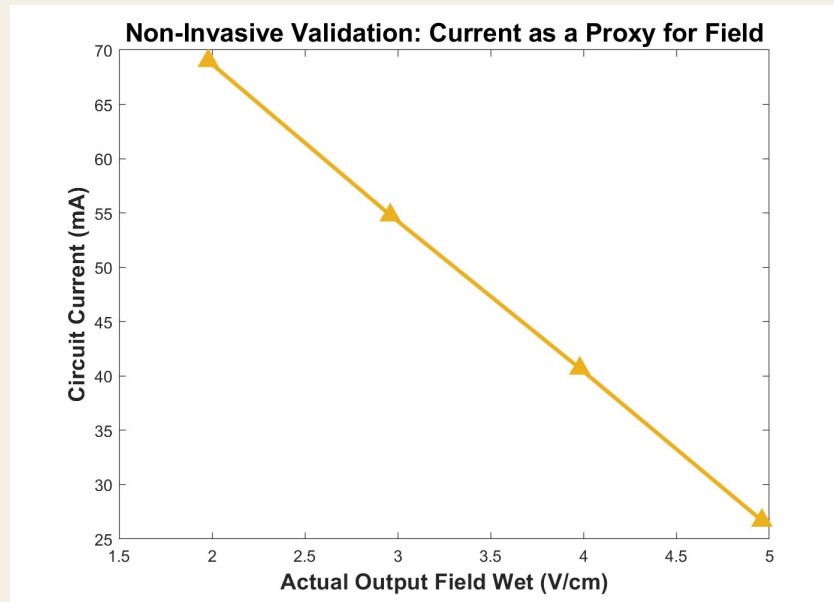


Figure 12.b - Measured Current via the INA vs the Output Wet Field

Helmholtz Measured Values

Pos ition	1											2											3										
X	-27	-21	-15	-9	-3	0	3	9	15	21	27	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
Y	0	0	0	0	0	0	0	0	0	0	0	-27	-21	-15	-9	-3	0	3	9	15	21	27	0	0	0	0	0	0	0	0	0	0	
Z	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-13.	5	-9	-3	3	9	5	25	13.		
mT	22.	46.	51.	51.	51.	51.	51.	51.	50.	45.	24.	24.	40.	51.	51.	51.	51.	50.	43.	21.	34.	44.	50.	51.	50.	47.	36.	5	1	6	1	5	

Smith Chart

S11 Smith Chart

